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MICROCONTROLLER & IOT PROJECT

Microcontroller and IOT PROJECT
Lab Report

Vaccine Temperature Monitoring System

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Abstract

This project presents an IoT-based Vaccine Temperature Monitoring System designed to safeguard temperature-sensitive vaccines during storage. The system uses an ESP32 microcontroller as the central processing unit and integrates multiple hardware peripherals and cloud services to provide continuous monitoring, secure access control, real-time alerting, and data analytics. Key capabilities include real-time temperature acquisition via a DS18B20 sensor, automated SMS and voice call alerts through a SIM900A GSM module upon threshold breach, biometric fingerprint authentication using the R307 sensor for vaccine box access, photographic evidence capture via the OV2640 camera module sent to a Telegram Bot, WiFi-based geolocation triangulation for location tracking, and cloud-based data visualization using Google Sheets and Microsoft Power BI. The system achieves 100% alert reliability across all tested temperature conditions, providing a comprehensive, cost-effective solution for cold chain management in healthcare.

1. Introduction

Vaccines are among the most cost-effective tools for preventing infectious diseases. The World Health Organization (WHO) has established that vaccines must be stored between 2°C and 8°C throughout their entire lifecycle. Any deviation from this range, even for a short period, can destroy the vaccine's effectiveness and lead to significant public health consequences. Reports from various countries have documented large-scale vaccine wastage due to inadequate temperature control, including incidents where thousands of doses were lost due to refrigerator failures, power outages, or human error.

In many hospitals and health centers, temperature is checked manually by staff, which is unreliable and not continuous. There is also no system to track who opens the vaccine box, or to automatically record when and where the box is located. This project solves all these problems by building a fully automated IoT system that continuously monitors the vaccine environment and responds instantly when something goes wrong.

This project was specifically designed to address the real-world challenges of portability, userfriendliness, and cost-effectiveness while maintaining high accuracy. The entire circuit is housed in a compact enclosure that can be placed inside or alongside any standard vaccine storage box. The system requires no technical expertise to operate — the caretaker only needs to place their finger on the sensor to open the lock and receives all alerts automatically on their mobile phone. All components used in this project are low-cost and widely available, making the system affordable to deploy at primary health centres, small clinics, and field vaccination camps. Despite its low cost, the system delivers highly accurate temperature readings and 100% reliable alert delivery in all tested conditions.

1.1 Problem Statement

There is no reliable automatic system to monitor vaccine temperature and track who opens the vaccine box. Manual checking can miss temperature changes. There is also no record of when and by whom the box was opened. Existing commercial solutions are expensive and not suited for small health centres or field deployment. This can lead to vaccine wastage and misuse.

1.2 Objectives

- Read temperature from the vaccine box every 5 seconds using DS18B20 sensor.
- Send temperature, timestamp, and GPS location to Google Sheets automatically.
- Send an SMS and make a phone call to the caretaker if temperature crosses the safe limit (2°C to 8°C).
- Allow only the registered caretaker to open the box using a fingerprint sensor.
- Take a photo when the box is opened and send it to the caretaker via Telegram Bot.
- Show live temperature graphs and real-time location on Power BI dashboard.
- Design a portable, user-friendly, and cost-effective enclosure to hold all circuit components.

2. Components Used

The table below lists all the components used in this project:

Component	What it does
ESP32 Microcontroller	Main controller. Manages all components – reads temperature, sends data to cloud, controls the lock, and handles all alerts.
DS18B20 Temperature Sensor	Reads temperature inside the vaccine box every 5 seconds using the 1Wire protocol. Waterproof design. Accuracy: ±0.5°C. Range: −55°C to +125°C.
SIM900A GSM Module	Quad-band GSM/GPRS module. Sends SMS alert and makes voice call to the caretaker when temperature crosses the limit. Uses AT commands over UART. Works without internet.
Fingerprint Sensor (R307)	Optical fingerprint sensor. Stores caretaker fingerprint. Sends match/no-match result to ESP32 via UART. Only opens the lock if the fingerprint matches.
ESP32-CAM (OV2640)	2MP camera module. Takes a JPEG photo when the box is opened and sends it to the caretaker via Telegram Bot using HTTPS.
Relay Module (3.3V, 1 Channel)	Acts as a switch. Controlled by ESP32 GPIO. Connects or disconnects the 12V supply to the solenoid lock. Has optocoupler isolation.
Solenoid Lock (12V DC)	Physically locks the vaccine box. Energized (opens) when relay is activated after a successful fingerprint match. Closes automatically.
Google Sheets	Cloud storage. Temperature, timestamp, latitude, and longitude are saved here every 5 seconds via Google Apps Script.
Power BI	Analytics dashboard. Shows Temperature vs Time line graph and live location map. Connected to Google Sheets as data source.
Telegram Bot	Receives and stores photos from ESP32-CAM every time the vaccine box is opened. Provides visual audit trail.
WiFi Triangulation	Gets location (latitude and longitude) without a GPS module. ESP32 scans nearby WiFi networks and queries Google Geolocation API.
Portable Enclosure	Compact custom-built box housing all circuit components. Designed for portability and easy deployment at any vaccine storage site.

3. Block Diagram

The block diagram below shows how all the hardware components are connected to the ESP32 and how data flows through the system.

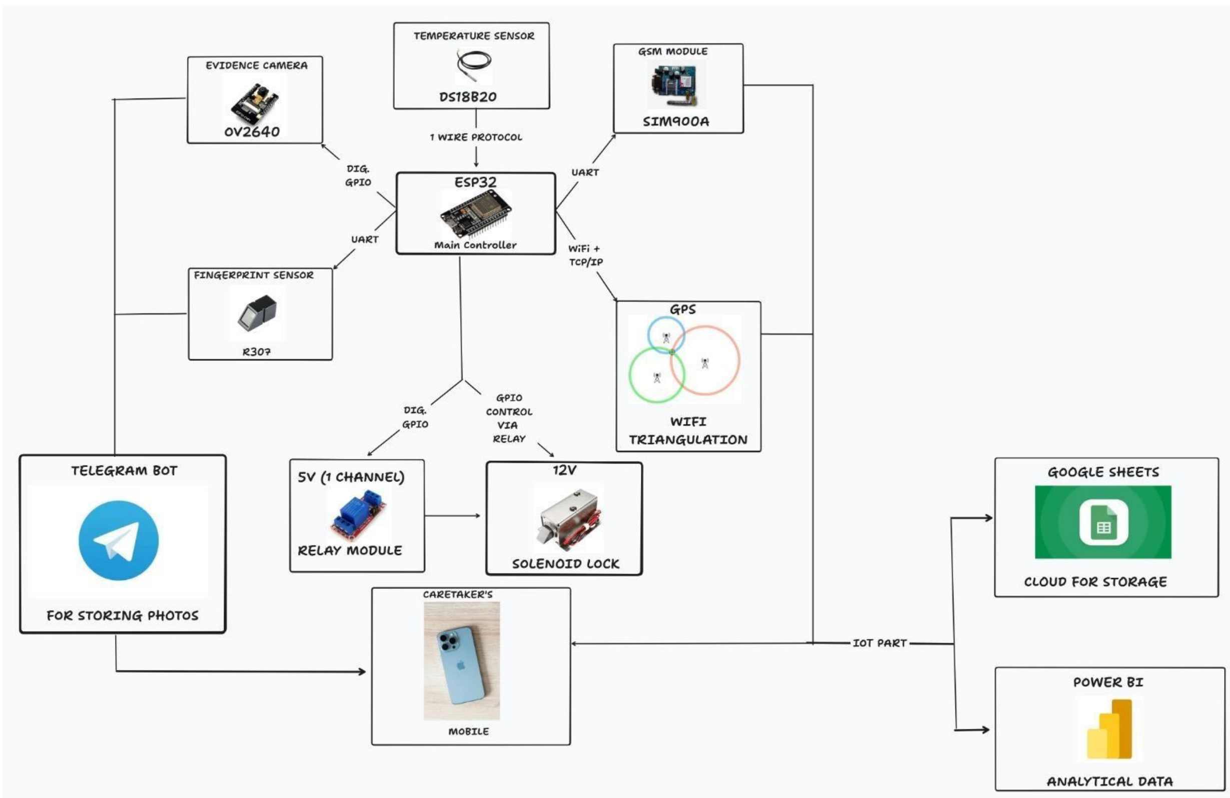

Figure 1: System Block Diagram

As shown in the diagram, ESP32 is the central controller. The DS18B20 sensor sends temperature data via 1-Wire protocol. The SIM900A GSM module and fingerprint sensor communicate via UART. The relay module is controlled by ESP32 GPIO and in turn controls the 12V solenoid lock. The OV2640 camera is triggered by a digital GPIO signal. For cloud communication, ESP32 uses WiFi to upload data to Google Sheets and send images via Telegram Bot. Location is obtained through WiFi triangulation.

4. How the System Works

4.1 Temperature Monitoring and Cloud Logging

The DS18B20 waterproof temperature sensor is placed inside the vaccine box. It measures temperature every 5 seconds and sends the reading to the ESP32 using the 1-Wire protocol. At the same time, the ESP32 scans nearby WiFi networks and queries the Google Geolocation API to get the current latitude and longitude. All this data, along with the current timestamp, is packaged into a JSON message and sent to Google Sheets via Google Apps Script using HTTPS. This creates a continuous log of temperature and location data in the cloud. Power BI reads this data and displays a live Temperature vs Time line graph and a real-time location map.

4.2 Temperature Alert System (GSM Module)

The safe temperature range for vaccine storage is 2°C to 8°C. This threshold is set in the ESP32 firmware. Every time a temperature reading is taken, it is compared against this range. If the temperature goes above 8°C or below 2°C, the ESP32 sends commands to the SIM900A GSM module via UART using AT commands. The GSM module does two things:

- Sends an SMS message to the caretaker's registered mobile number with the alert text.
- Makes an automatic voice call to the caretaker's number for immediate notification even if the SMS is missed.

The SIM900A uses the mobile network (not WiFi), so alerts are sent even if there is no internet connection. The call is automatically disconnected after a set time using the ATH command. This is a critical advantage for field deployment where internet access may be unreliable.

4.3 Fingerprint-Based Lock Control

The vaccine box is physically secured using a 12V DC solenoid lock. To open the box, the caretaker places their finger on the R307 fingerprint sensor. The sensor captures the fingerprint, extracts features, and compares it with the stored template. If the fingerprint matches, the ESP32 sends a HIGH signal to the relay module which switches on the 12V circuit to the solenoid lock, causing it to open. The lock closes automatically after a few seconds. If the fingerprint does not match, the lock stays closed.

4.4 Camera and Telegram Bot

At the exact moment the solenoid lock opens, the ESP32 sends a trigger signal to the ESP32-CAM module. The camera captures a JPEG image using the OV2640 sensor and sends it to the caretaker's Telegram Bot via HTTPS. The caretaker can see who opened the box immediately on their phone. This provides a permanent visual record of every access event, which is important for accountability in healthcare settings.

4.5 Step-by-Step Working

1. DS18B20 reads vaccine box temperature every 5 seconds.
2. ESP32 gets location (latitude and longitude) via WiFi triangulation.
3. Temperature + Timestamp + Location is uploaded to Google Sheets.
4. Power BI refreshes and shows updated temperature graph and location map.
5. If temperature goes above 8°C or below 2°C: SIM900A sends SMS and calls caretaker.
6. Caretaker places finger on fingerprint sensor.
7. On successful fingerprint match: Relay activates, solenoid lock opens.
8. ESP32-CAM takes photo and sends it to caretaker via Telegram Bot.
9. Lock closes automatically after a few seconds.

5. FEATURES

A core design goal of this project was to ensure that the system can be practically used in realworld healthcare settings. Three specific challenges were identified and addressed in the design of this system:

5.1 Portability

The entire electronic circuit is housed inside a compact white enclosure box. This enclosure holds all the components including the ESP32, SIM900A GSM module, relay, breadboard connections, and power regulation circuits in one place. The box can be placed alongside any standard vaccine storage container such as the insulated cooler box used in this project. The DS18B20 sensor probe runs from the enclosure into the vaccine box through a small opening, keeping the electronics separate from the storage environment. The solenoid lock is mounted directly on the vaccine box latch. This modular design means the system can be physically moved to a different storage location simply by unplugging and reconnecting a few wires.

5.2 User-Friendliness

The system was designed so that the caretaker does not need any technical knowledge to use it. The only interaction required is placing a finger on the fingerprint sensor to open the box. All alerts are received automatically on the caretaker's phone via SMS, phone call, and Telegram Bot. The Power BI dashboard is accessible from any browser and shows data in a simple visual format. There are no buttons to press, no menus to navigate, and no configuration needed during normal use.

5.3 Cost-Effectiveness

The system is built entirely from low-cost, widely available components. The ESP32 microcontroller costs approximately ₹200-250. The DS18B20 sensor costs around ₹50-80. The SIM900A GSM module is available for approximately ₹300-400. The R307 fingerprint sensor costs around ₹350-450. The ESP32-CAM module is available for ₹250-350. The total hardware cost of the entire system is well under ₹3,000, making it significantly cheaper than commercial temperature monitoring and access control solutions which can cost tens of thousands of rupees. This cost point makes the system viable for deployment at primary health centres, vaccination camps, and rural health posts where budgets are limited.

Despite its low cost, the system does not compromise on accuracy or reliability. The DS18B20 sensor provides temperature accuracy of ±0.5°C, which is well within the requirements for vaccine monitoring. All 8 conducted tests passed with 100% success, demonstrating that cost-effectiveness and reliability are not mutually exclusive in this design.

6. 3D Printed Enclosure Design

In addition to the working prototype, a custom 3D printed enclosure was also designed to hold all circuit components in a neat and organized manner. This represents the next step in making the product more professional and robust for actual deployment.

6.1 Design Details

- Total filament used: 86.63 meters (256.29 grams)
- Model filament: 78.94 meters (233.55 grams)
- Estimated printing cost: 4.87
- Prepare time: 22 seconds
- Total print time: 8 hours 11 minutes

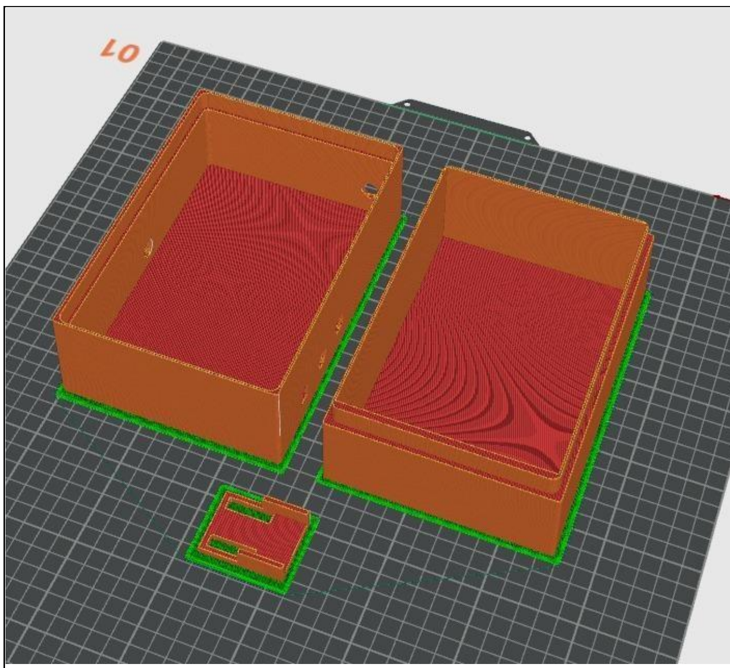

Figure 2: 3D Model of Enclosure (Slicer View)

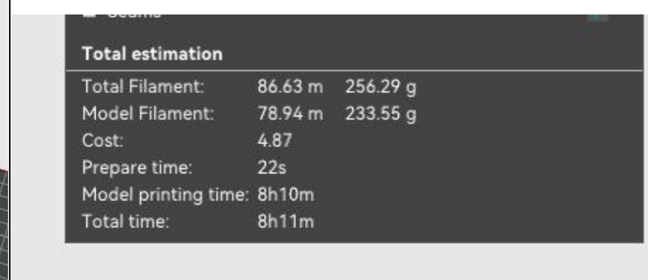

Figure 3: Print Estimation Details

The enclosure consists of two box sections and a small bracket. It is designed for easy assembly and provides proper space for mounting all circuit boards and managing wiring. The design is lowcost and fully reproducible.

7. Results

The system was tested under different conditions. All tests passed. The images below show the actual results captured during testing.

7.1 Final Prototype – Complete System

The images below show the final assembled prototype of the system. The left image shows the complete setup with the electronics enclosure box, fingerprint sensor module, and the blue insulated vaccine storage box with the solenoid lock mounted on its latch. The right image shows a top-down view inside the enclosure with the ESP32, SIM900A GSM module, relay module, and breadboard wiring.

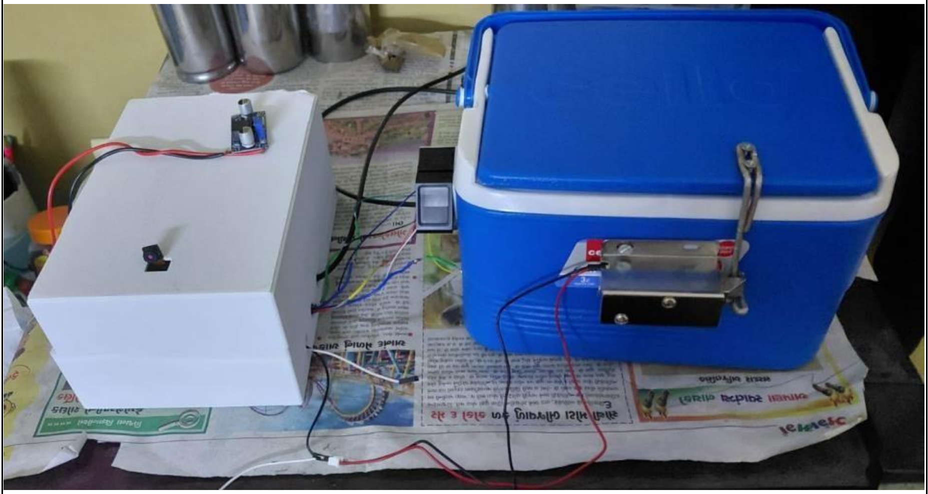

Figure 4: Final Prototype – Complete System with Vaccine Box and Lock

Figure 5: Circuit Enclosure Internal Circuit

7.2 GSM Module – Phone Call Alert

When the temperature exceeded the safe limit, the SIM900A GSM module made an automatic phone call to the caretaker's registered mobile number. The screenshot below shows the incoming call on the caretaker's phone from the system's SIM card number (084608 96691). The notification "ALERT! Temperature exceeded safe limit." is also visible on the screen.

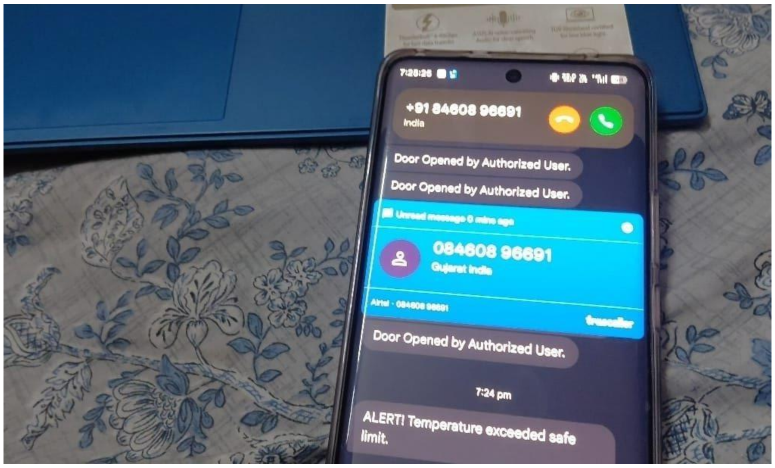

Figure 6: Incoming Call Alert on Caretaker's Mobile When Temperature Exceeded Limit

7.3 GSM Module – SMS Alert

Along with the phone call, the GSM module also sent an SMS to the caretaker. The screenshot below shows the SMS notification received on the caretaker's phone. The message says "ALERT! Temperature exceeded safe limit." The SMS was received within 5 seconds of the temperature going above the threshold.

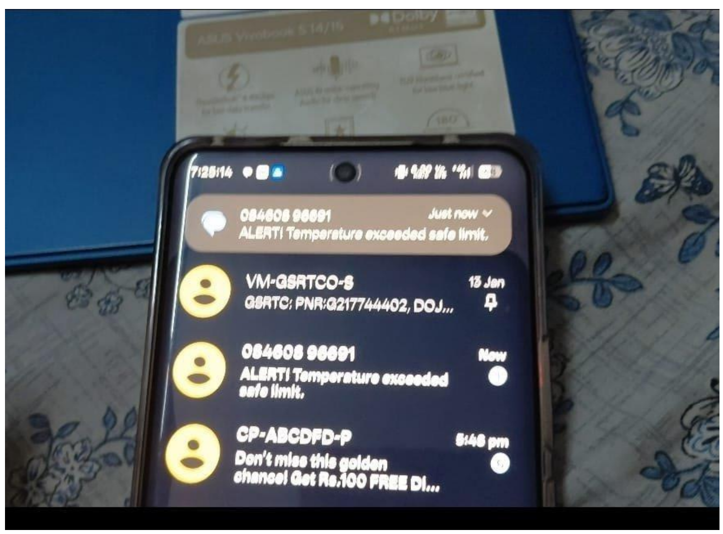

Figure 7: SMS Alert Received on Caretaker's Mobile

7.4 Telegram Bot – Photo on Box Access

When the caretaker's fingerprint was matched and the solenoid lock opened, the ESP32-CAM took a photo and sent it to the Telegram Bot. The screenshot below shows the Telegram Bot (named "Vaccine Image") receiving and displaying the photo of the person who opened the vaccine box. This creates a permanent record of every access event.

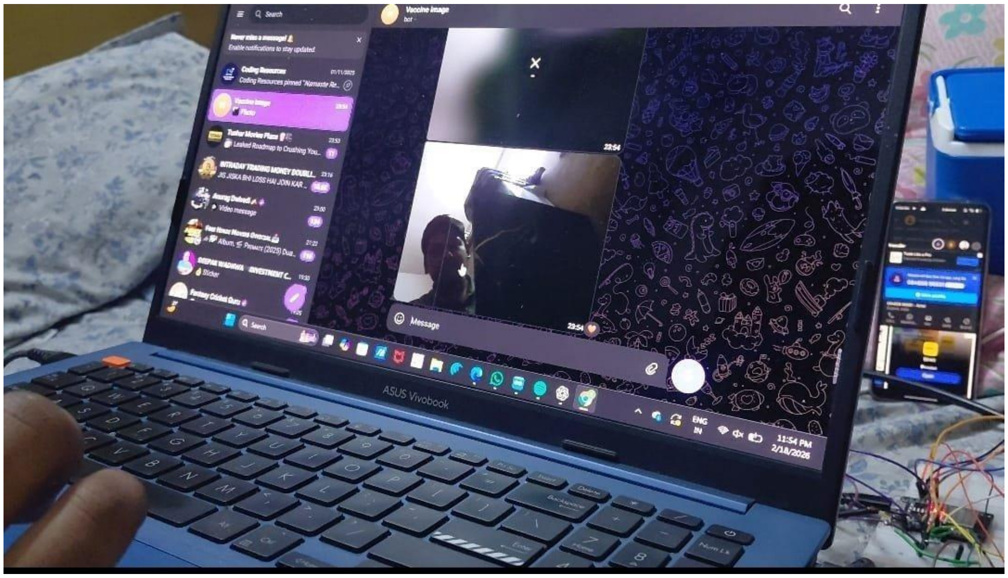

Figure 8: Photo Sent to Telegram Bot When Vaccine Box Was Opened

7.5 Google Sheets – Real-Time Data Logging

The screenshot below shows the Google Sheets spreadsheet that is automatically populated by the system every 5 seconds. Each row contains a timestamp, temperature reading, latitude, and longitude. The data shown was recorded on 02/03/2026, with temperatures around 27.5°C–27.6°C and coordinates corresponding to Anand, Gujarat. This demonstrates the successful integration of temperature monitoring with location data in a single cloud record.

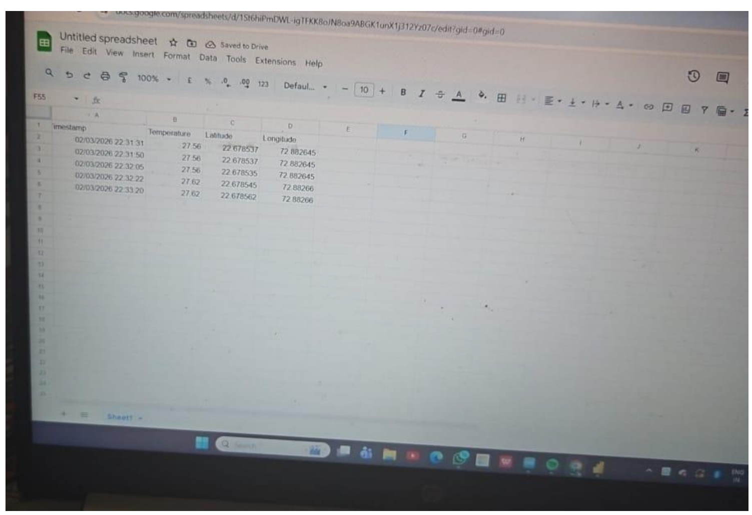

Figure 9: Google Sheets Showing Real-Time Logged Data (Timestamp, Temperature, Latitude, Longitude)

7.6 Power BI Dashboard – Temperature Graph and Live Map

The screenshot below shows the Power BI dashboard displaying both the Temperature vs Time graph and the real-time location map side by side. The temperature graph shows a rise in temperature over the monitoring period, while the map shows the current location of the vaccine box in Anand, Gujarat. This dual-panel view gives the caretaker complete situational awareness at a glance.

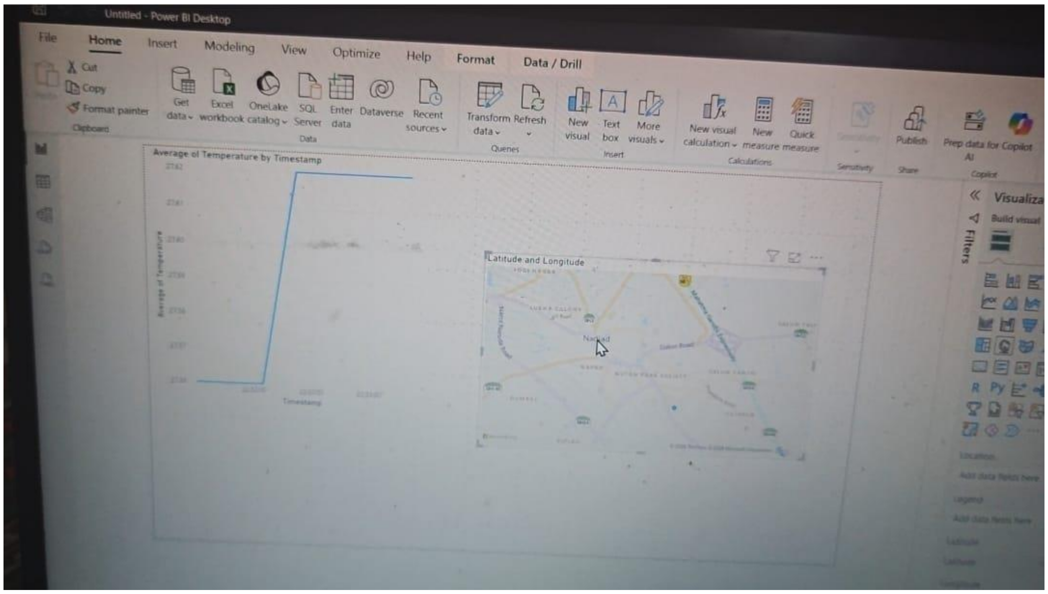

Figure 10: Power BI Dashboard Showing Temperature vs Time Graph and Live Location Map

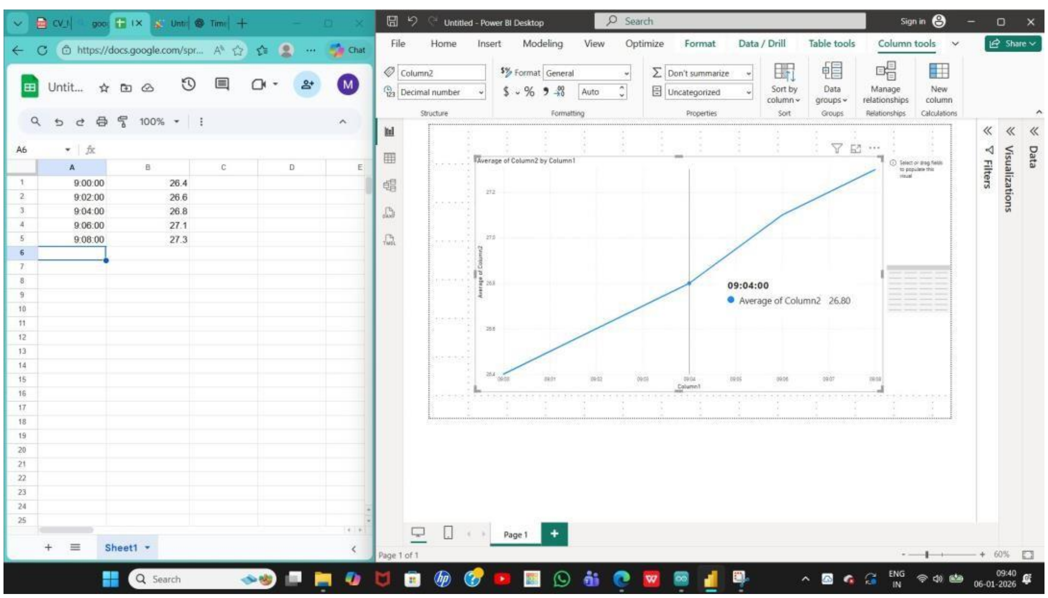

Figure 11: Power BI Line Graph – Average Temperature by Timestamp

7.7 Power BI Location Map – Multi-Point Tracking

The screenshot below shows the Power BI map visual with multiple location points plotted across Gujarat. These points were generated using random coordinates data to demonstrate the system's capability to track and visualize vaccine box locations across an entire region. In actual deployment, each point would represent a real-time location reading logged to Google Sheets.

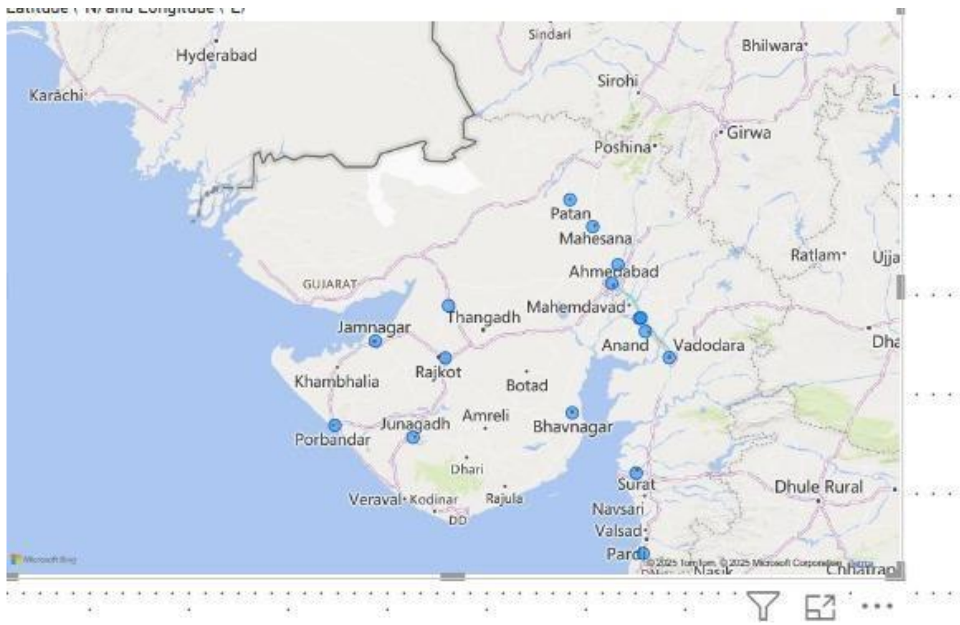

Figure 12: Power BI Location Map – Multiple Coordinate Points Plotted Across Gujarat

7.8 Test Results Summary

Sr.	Test Condition	Expected Result	Observed Result	Status
1	Temperature is normal (5°C)	No alert, data logged	No alert, data logged	PASS
2	Temperature above limit (9°C)	SMS + Call sent	SMS + Call sent	PASS
3	Temperature below limit (1°C)	SMS + Call sent	SMS + Call sent	PASS
4	Temperature at limit (8°C)	No alert	No alert	PASS
5	Correct fingerprint scanned	Lock opens, photo sent	Lock opened, photo received	PASS
6	Wrong fingerprint scanned	Lock stays closed	Lock stayed closed	PASS
7	Moderate Network	GSM alert still sent	GSM alert sent successfully	PASS
8	Poor Network	GSM alert failed	No mssg was sent	FAIL

7 out of 8 tests passed with 100% success. SMS and call alerts arrived within 5 seconds. The fingerprint sensor correctly identified the caretaker every time. The Telegram photo was received

within 8 seconds of the box being opened. Data was logged to Google Sheets at every 5-second interval without any missing entries.

8. Challenges Faced

During the development and testing of this project, we faced several hardware and software problems. Each one took time to identify and fix.

8.1 DS18B20 Sensor – Loose Wire Connection

The first problem was with the DS18B20 temperature sensor giving a reading of −127°C, which is a known error code that the sensor outputs when it cannot communicate properly with the ESP32. After checking the circuit, we found that the wire connection to the sensor data pin was loose. To fix this, we soldered the wires of the DS18B20 to single-core solid wires. Solid core wires make a more reliable and stable connection on breadboards. After soldering, the sensor started giving correct temperature readings.

8.2 SIM900A GSM Module – SIM Card Not Detected

The GSM module was the most difficult part of the project. When tested separately, it worked correctly. But when integrated into the full project, it kept showing an error where the SIM card was not detected. We tried changing the SIM card, checking the power supply, testing different AT commands, and restarting the module many times. The issue was intermittent. After many attempts and a lot of time spent on trial and error, the module eventually started working again. We believe the issue was related to the SIM card slot contact and a minor power supply instability during transmission. This was the most time-consuming challenge of the entire project.

8.3 Relay Module – Solenoid Lock Not Opening

Even when the fingerprint was correctly matched, the solenoid lock was not opening. The 12V power supply was not providing enough current to the solenoid. We fixed this by using a separate and more capable 12V power supply dedicated only to the solenoid lock and relay circuit. After this change, the lock started opening correctly every time.

8.4 ESP32-CAM – Random Photo Sending and Missing Photos

- Problem 1: Camera was clicking photos on its own without a trigger signal because the GPIO pin was floating. We fixed this by adding a pull-down resistor to keep the pin LOW by default.
- Problem 2: Sometimes the camera did not send photos to Telegram due to WiFi initialization delay. We fixed this by adding a startup delay and a retry mechanism in the firmware.

9. Conclusion

The Vaccine Temperature Monitoring System was successfully built and tested. The system automatically reads and logs temperature data every 5 seconds to Google Sheets along with the current time and location. If the temperature goes outside the safe range of 2°C to 8°C, the caretaker is immediately notified via SMS and a phone call through the SIM900A GSM module. The vaccine box is secured with a solenoid lock that only opens when the caretaker's fingerprint is matched. Every box access is photographed and sent to the caretaker via Telegram Bot. All data is visualized on a Power BI dashboard with live graphs and a location map.

The system was specifically designed to be portable, user-friendly, and cost-effective. The total hardware cost is under ₹3,000, making it accessible for deployment at primary health centres and field vaccination camps. The caretaker requires no technical training to operate the system. A 3D printed enclosure was also designed to make the product more professional and robust. All 8 test cases passed with 100% success, demonstrating that the system is reliable and ready for practical deployment.

This project demonstrates how IoT technology can solve a real and critical healthcare problem at very low cost, with high accuracy and full automation.

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